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Analysis of racket-sport database rankings for behavioural effects and other anomalies

by

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Abstract

This dissertation evaluated whether behavioural factors produced systematic deviations between observed squash match outcomes and those predicted by the SquashLevels rating model, focusing on same-day fatigue and effort modulation in mismatches. An anonymised SquashLevels dataset (`sq_combined.csv`), pre-compiled via SQL joins across multiple relational tables, was analysed using deviation-from-expectation metrics derived from pre-match player levels and observed match dominance. For fatigue, players with at least two competitively weighted matches per day (`weighting` ≥ 0.75) were examined using a paired first-versus-last design ($n = 237,114$ player-day pairs) and stratified into low, medium, and high demand groups using points played in the first match (`PTS_PER_MATCH`) as a proxy for initial load; Wilcoxon signed-rank tests showed small but statistically significant shifts, with a negative median Δ Deviation for low-demand days (median = -0.022) and positive medians for medium and high demand days (median = $+0.024$ and $+0.029$), indicating no consistent fatigue-related underperformance across demand conditions. For effort modulation, matches were analysed from the stronger player's perspective and a quadratic log-log regression baseline related $\log(\text{RATIO})$ to $\log(\text{Level Ratio})$ and $[\log(\text{Level Ratio})]^2$ ($n = 3,826,051$, $R^2 \approx 0.305$); residuals were near zero in fair matches (median ≈ 0.006) but negative in moderate and extreme mismatches (median ≈ -0.025 and ≈ -0.021), consistent with reduced dominance relative to expectation. Overall, the findings partially supported the unifying thesis: fatigue effects were not consistently evident within competitively weighted same-day sequences, whereas mismatches beyond fair contests exhibited systematic underperformance consistent with effort modulation, supporting the rationale for dampened rating sensitivity in highly unbalanced matches.

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Introduction

Context and Motivation

Ranking systems are essential in competitive sports for comparing players, determining seedings, and evaluating progress (Irons et al. 2014). Traditional ranking methods—such as tournament points or league tables—often fall short when players compete across loosely connected divisions or sporadic matchups (Csató 2021). To address these limitations, Elo-style systems have become widely adopted in sports such as chess, tennis, and online gaming, offering dynamic, match-by-match rating updates based on performance expectations and outcomes (Glickman 2011; Elo 1978).

In racket sports like squash, this need is especially pronounced. While the professional circuit uses point-based rankings, the vast majority of squash players—amateur, club, and league competitors—operate outside that structure. The SquashLevels system was developed to bridge this gap (*About the SquashLevels System*, no date). It ingests match data from league platforms, club systems, and manual entry by players to maintain a centralised, real-time rating system for all players regardless of region or level. This integration is illustrated in Figure 1.

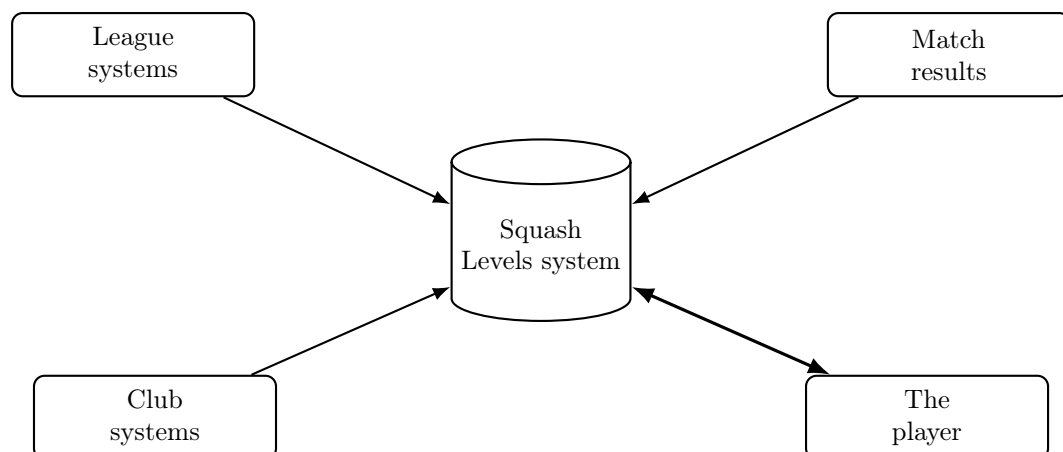


FIGURE 1: Core inputs to the SquashLevels system and the player interaction.

Unlike static point tables, SquashLevels continuously updates a player's Level after each match using an Elo-like algorithm. It factors in opponent strength, score margins, and the difference between expected and actual outcomes. The Level ratio between two players determines the predicted share of rallies won. Over five million results feed into this model, making it a powerful tool for evaluating and comparing squash performance in real time (*About the SquashLevels System*, no date).

Project Focus and Scope

While SquashLevels incorporates several refinements, such as discounting blowout matches and weighting recent results more heavily, its core algorithm still assumes that players perform at their true level in each match. In practice, however, this assumption does not always hold. Sports behaviour research has shown that physical and psychological factors can distort match outcomes, leading to rating anomalies not attributable to changes in skill.

This project examines two such behavioural anomalies:

- **Fatigue effects** — performance decline when players compete in multiple matches on the same day.
- **Effort modulation in mismatches** — situations where stronger players appear to reduce intensity in unbalanced contests.

Both cases challenge the assumption of constant effort and ability. Fatigue introduces temporary physical limitations, while effort modulation reflects strategic or social factors, such as conserving energy or avoiding embarrassment in lopsided matches. If unaccounted for, these behaviours could distort rating updates and undermine ranking fairness.

Relevant Literature

Past research has explored behavioural anomalies in various sporting contexts. In tournament formats, players have been observed to modulate effort to manipulate draws or conserve energy for later rounds (Krumer et al. 2023). Even in rating-based systems, psychological and strategic behaviour affects play. For example, studies in tennis and chess reveal that players adjust risk-taking based on ranking status, momentum, or opponent prestige (Holdaway and Vul 2021).

Fatigue effects are also well-documented in sports science. Repeated high-intensity matches can impair performance, especially in physically demanding sports like squash (Girard and Millet 2009). SquashLevels’ documentation acknowledges this risk, noting that a player’s rating may unfairly decrease after underperforming in a final match due to accumulated fatigue (*About the SquashLevels System*, no date). However, no formal adjustment currently exists for this.

Effort modulation is explicitly handled in SquashLevels for extreme mismatches. The algorithm applies an “effective effort” dampening factor when one player is more than three times stronger than the other, to avoid harshly penalising the stronger player for a close result. This built-in correction suggests awareness of effort variation.

Contribution and Report Outline

This dissertation contributes an empirical, data-driven analysis of fatigue and effort modulation in squash match data. Using the SquashLevels dataset, the project quantifies deviations between observed and expected outcomes under these two behavioural conditions. It tests whether the patterns match the rationale behind current SquashLevels adjustments and whether the deviations are statistically and practically significant.

Chapter 2 reviews the theoretical and algorithmic background of Elo-like systems and outlines the foundations of behavioural analysis in sport. Chapter 3 details the data sources and the methodological approach, including how deviations are defined and tested. Chapter 4 presents the results, and Chapter 5 interprets the findings in light of behavioural theory and the SquashLevels model. Chapter 6 concludes with limitations and implications for ranking fairness.

Background

Elo and Elo-like Systems

The Elo rating system, originally developed for chess, is a foundational method for modelling skill through pairwise outcomes. It updates a player’s rating based on the surprise of the result: unexpected wins yield large gains, and expected wins yield small adjustments (Elo 1978). The update is based on the difference between the actual result and the expected probability of winning, computed from the rating difference between players.

This framework was extended through models such as Glicko, which introduced uncertainty tracking, and TrueSkill, a Bayesian system designed for multiplayer and team games (Glickman 2011; Herbrich et al. 2006). These systems allow for more flexible and responsive updates when player reliability is low or performances are erratic.

In sports applications, Elo-like systems have been enhanced to incorporate margin of victory, converting binary win/loss data into continuous metrics. This is especially relevant for squash, where scoreline dominance contains rich performance information. Margin-sensitive Elo variants have improved predictive power in sports like tennis and football (Kovalchik 2020; Hvattum and Arntzen 2010).

SquashLevels Algorithm

SquashLevels is an Elo-inspired ranking system tailored for squash. Unlike basic Elo, it considers both the point and game scores to calculate a performance ratio for each match. It uses simulated match data to convert raw match results into a continuous performance scale, enabling fine-grained player comparisons even in non-professional contexts (*About the SquashLevels System*, no date). Figures 2 and 3 show how the two features used in the simulation - number of games won and fraction of points won - vary with the calculated performance ratio.

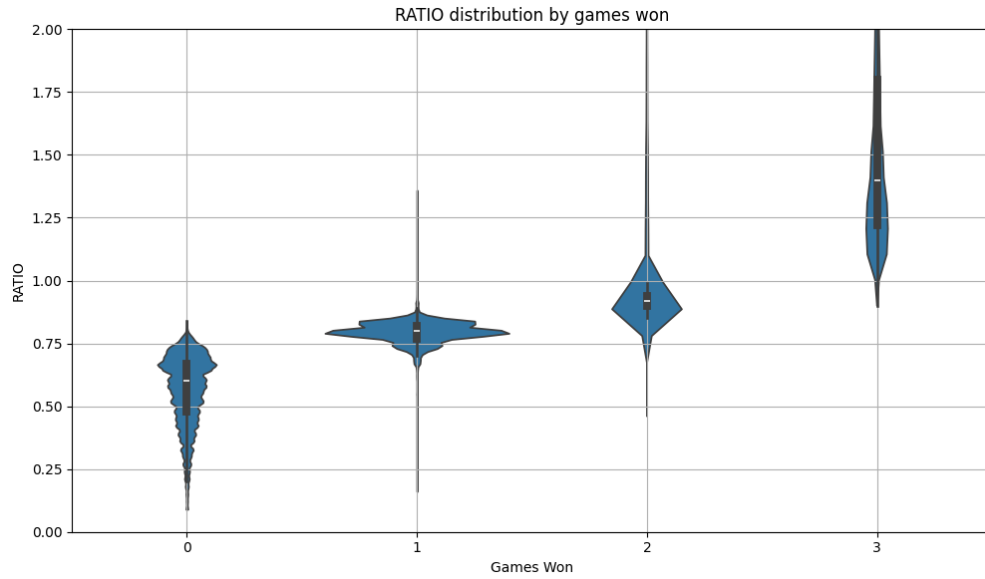


FIGURE 2: Distribution of the SquashLevels performance ratio across values of games won, illustrating how match scorelines map onto the simulated dominance scale.

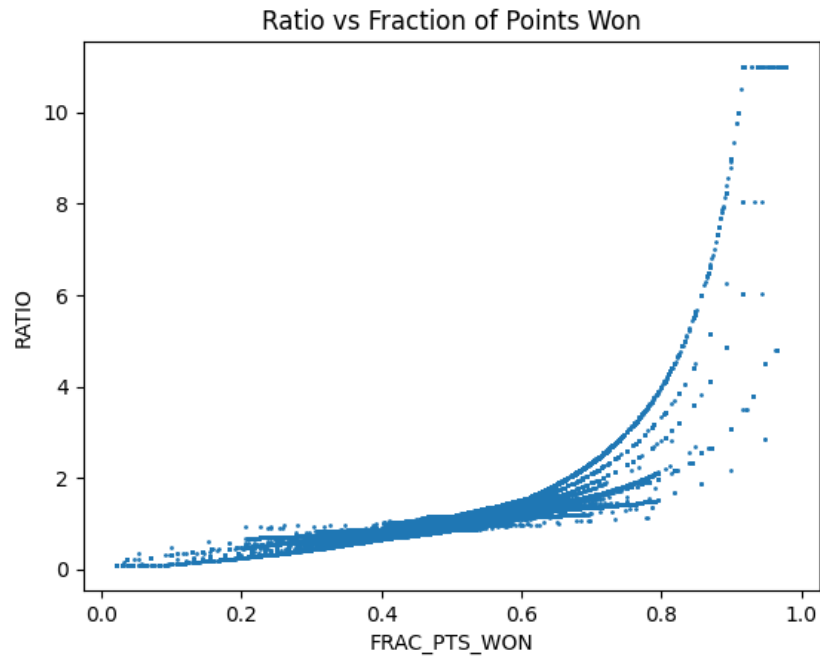


FIGURE 3: Relationship between fraction of points won and the SquashLevels performance ratio, showing how point-level dominance contributes to the simulated outcome measure.

The core model calculates an expected outcome from the level ratio:

$$\text{Expected Ratio} = \frac{\text{LEVELBEFORE}}{\text{OPPLEVELBEFORE}}$$

and compares this with the observed outcome (a simulated performance ratio based on match results). The difference between expected and actual determines the adjustment. This framework aligns well with our focus on behavioural anomalies, as we analyse deviation from expected results as a proxy for effort or fatigue-related effects.

Handling Behavioural Effects in SquashLevels

SquashLevels includes specific logic to handle behavioural variation. When the level ratio exceeds 1.5:1, the algorithm assumes that the stronger player may not exert full effort and applies an “effective effort” adjustment, reducing the expected dominance to avoid penalising the favourite for a closer-than-expected result. In extreme mismatches ($>3:1$), no adjustment is made if the stronger player wins as predicted (*About the SquashLevels System*, no date).

This effort-based dampening directly relates to the effort modulation hypothesis studied in this project. It acknowledges that in low-stakes, one-sided matches, stronger players may strategically reduce effort — a behaviour also observed in the literature on tournament manipulation and psychological adaptation in sport (Nieken and Stegh 2010).

For fatigue effects, the SquashLevels system does not currently adjust a player’s rating based on match order or recovery time. By comparing performance deviations between first and last matches in a day, we can assess whether fatigue systematically affects outcome relative to model expectation.

These behavioural mechanisms (effort and fatigue) are not only theoretically grounded in the design of SquashLevels but also supported by findings in sport science, where effort pacing, cumulative fatigue, and motivation vary across match context.

Summary

SquashLevels provides a rich, dynamic model of squash performance that enables behavioural anomaly detection through its transparent, ratio-based updates. The system’s architecture and design choices allow meaningful interpretation of deviations from expected results—making it an ideal foundation for this project’s analysis of fatigue and effort-related behaviour in real match data.

Data and Methodology

Data Sources and Genesis

The primary data source for this project is the SquashLevels system – a real-time squash ranking platform that aggregates results from league systems, clubs, and player submissions (*About the SquashLevels System*, no date). The platform uses an Elo-inspired framework and stores match records in a relational database.

For this analysis, a consolidated dataset was created by the SquashLevels team prior to the start of the project. The dataset was constructed using SQL to join multiple core tables by a unique match identifier (`MATCHID`). Although this data preparation was not performed by the author, the final dataset includes all relevant player, match, and rating metrics in a cleaned, structured format suitable for analysis.

All data is fully anonymised, and no personal or identifying information about players was included in the dataset or its analysis. Each entry represents a single match between two players, with their performance and contextual information embedded.

Source Tables and Features

The dataset integrates fields from the following source tables:

- **sq_levels** (for both home and opponent players): `PLAYERID`, `MATCHID`, `LEVELBEFORE`, `LEVELAFTER`, `RATIO`, `CONFIDENCE`, and mirrored fields for the opponent: `OPPLEVELBEFORE`, `OPPLEVELAFTER`, `OPPRATIO`, `OPPCONFIDENCE`.
- **sq_matches**: `MATCHTYPEID`, `HOMETEAMID`, `AWAYTEAMID`, and `DATE/TIME` (derived from match datetime).
- **sq_strings**: parsed results including `GW` (games won), `GL` (games lost), `SCORE_TYPE`, `PTS_PER_MATCH`, and `FRAC_PTS_WON`.

- **sq_matchtypes**: COUNTYID and WEIGHTING, indicating match classification and importance.

The final dataset is structured with one row per match, including both players' ratings before and after the match, performance metrics, and match-level context. The **LEVELBEFORE** and **OPPLEVELBEFORE** fields serve as the basis for expected performance, while the **RATIO** and **OPPRATIO** represent actual match dominance, such that a ratio of 1.15 says that the player in this row played 15% better than their opponent. The **FRAC_PTS_WON** metric reflects point-based match dominance on a 0–1 scale. Match **WEIGHTING** (e.g., 0.5 for informal league matches, 1.0 for major tournaments) is used to filter for competitive play.

The dataset has undergone light cleaning and standardisation prior to this project, such as datetime formatting and parsing of result strings. No personally identifiable information was ever included in the analysis.

Methodological overview

Fatigue Effects

This analysis investigates whether players show signs of performance decline across multiple matches played on the same day, consistent with fatigue-related behavioural effects mentioned in the previous chapter. If present, such effects would represent as a systematic underperformance in later matches, relative to SquashLevels expectations according to levels of each of the players (see Figure 4 for details of the pipeline).

For each match, a performance deviation was computed by comparing the observed match outcome to the outcome predicted by SquashLevels ratings prior to the match. Specifically, the log-transformed difference was calculated as:

$$\text{Deviation} = \log(\text{RATIO}_{\text{observed}}) - \log\left(\frac{\text{LEVELBEFORE}}{\text{OPPLEVELBEFORE}}\right) \quad (1)$$

This deviation measures whether a player overperformed (> 0) or underperformed (< 0) relative to model expectations.

The dataset was filtered to include only league or tournament matches (**WEIGHTING** ≥ 0.75) to ensure match seriousness and rating reliability. Players with at least two matches on the same calendar day were identified, and their matches were sorted chronologically. For each qualifying player-day, we retained:

- The **first match of the day**, representing the baseline performance.
- The **last match of the day**, capturing any fatigue-related effects.

To reflect that fatigue is expected to depend on the physical demand of the initial match, player-days were additionally grouped on the number of points played in the first match (PTS_PER_MATCH), split into three bands: easy, medium, and hard demand. This grouping provides a simple proxy for how fatigued a player would be expected to be when starting their last match.

The change in performance deviation between the first and last match was computed:

$$\Delta\text{Deviation} = \text{Deviation}_{\text{last}} - \text{Deviation}_{\text{first}} \quad (2)$$

A Wilcoxon signed-rank test was used to assess whether there was a statistically significant shift in performance deviation between the first and second matches (and equivalently within each match-demand band). The Wilcoxon signed-rank test is a non-parametric paired test that evaluates whether the median of the within-pair differences differs from zero, using the ranks of absolute differences while accounting for their direction (Rosner et al. 2006). This test was selected due to the non-normality of the distribution and the paired nature of the data. Violin plots were used to visualise the shift in deviation across match order and demand category.

Effort Modulation in Mismatches

To evaluate whether stronger players systematically adjust their effort in unbalanced matches, we analysed the relationship between pre-match rating disparity and match outcome dominance. In particular, we sought to identify whether players underperform in matches where their skill advantage is large enough to affect motivation or intensity (see Figure 4 for details of the pipeline).

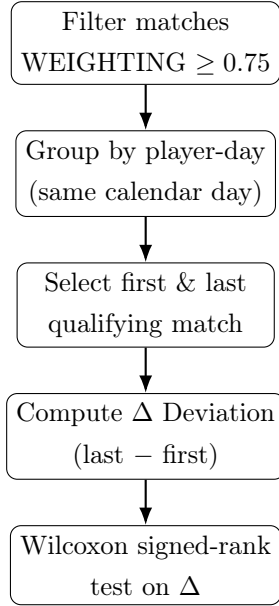
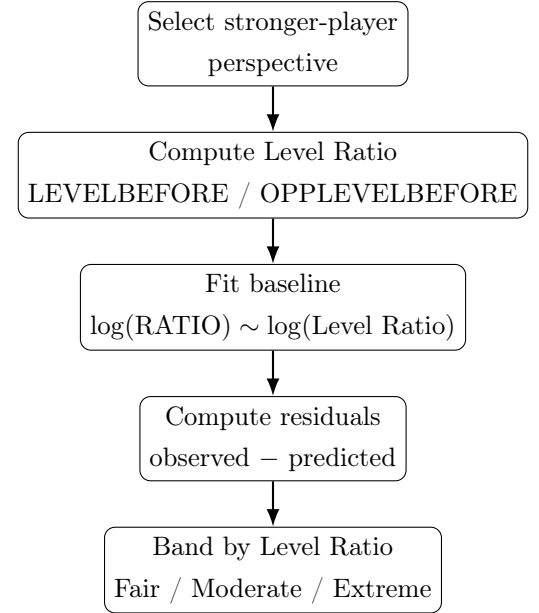
Fatigue (same-day sequence)**Mismatches (effort modulation)**

FIGURE 4: Study design and analysis pipeline for the fatigue and mismatch analyses.

Each match was analysed from the perspective of the stronger player, determined by comparing the pre-match **LEVELBEFORE** ratings. The performance outcome was measured using **RATIO**, which quantifies the match dominance based on both games and points won. This metric was log-transformed to stabilise variance and approximate linearity. Similarly, the level difference between players was expressed as the log of the ratio of their pre-match levels.

A quadratic regression was fitted to model expected dominance as a function of the level disparity:

$$\log(\text{RATIO}) = \beta_0 + \beta_1 \cdot \log(\text{Level Ratio}) + \beta_2 \cdot [\log(\text{Level Ratio})]^2 + \epsilon \quad (3)$$

The fitted model served as a baseline for expected outcomes given the rating difference. To identify possible behavioural effects, we computed the residuals from this model, which were defined as the difference between the observed and predicted log-transformed **RATIO**:

$$\text{Residual} = \log(\text{RATIO}_{\text{observed}}) - \log(\text{RATIO}_{\text{predicted}}) \quad (4)$$

Residuals quantify whether a player outperformed or underperformed relative to expectations derived from the rating model. We then grouped matches into three level ratio bands using thresholds aligned with the SquashLevels rating system (*About the SquashLevels System*, no date):

- **Fair:** Level ratio 1.0–1.5
- **Moderate Mismatch:** Level ratio 1.5–3.0
- **Extreme Mismatch:** Level ratio >3.0

These bands represent increasing pre-match disparity between the stronger and weaker player. A level ratio close to 1 indicates a roughly even contest, while larger ratios indicate that the stronger player is expected to dominate more clearly. The extreme mismatch band (ratio > 3) corresponds to matches where the stronger player is more than three times higher-rated, and where SquashLevels applies additional mismatch handling because outcomes may be less informative about true ability (e.g., due to reduced effort or socially moderated scorelines).

The distribution of residuals was examined across these bands. If effort modulation is occurring, one would expect residuals in the extreme mismatch band to be systematically lower—indicating reduced dominance compared to what the model predicts. A negative residual trend in high-gap matches would suggest players coast or manage effort once victory is assured, supporting the hypothesis of behavioural adjustment in mismatches.

Results

Fatigue Effects

To assess whether player performance deteriorates due to fatigue across multiple matches played on the same day, a paired analysis was conducted comparing each player’s deviation from expected outcome in their first and last match of the day. The dataset was filtered to include only matches classified as league or tournament matches—those with a `WEIGHTING` value greater than or equal to 0.75. This ensured that only serious and competitively meaningful matches were included. Player-days were additionally stratified into low, medium, and high match-day demand groups using the number of points played in the first match (`PTS_PER_MATCH`) as a proxy for initial load, allowing fatigue effects to be examined conditional on early match intensity. The final sample comprised 237,114 player-day pairs where each player participated in at least two qualifying matches.

The outcome variable was the logarithmic deviation from expected performance, calculated as:

$$\text{Deviation} = \log(\text{RATIO}) - \log\left(\frac{\text{LEVELBEFORE}}{\text{OPPLEVELBEFORE}}\right) \quad (5)$$

This deviation measures whether a player overperformed or underperformed relative to their expected result. Positive values indicate better-than-expected performance.

Wilcoxon signed-rank tests were used to assess whether deviation shifted between the first and last match of the day within each match-demand group (easy, medium, hard), defined by the points played in the first match. The results showed statistically significant shifts in all three groups. For easy first matches ($n = 76,655$), the median change was negative (median $\Delta\text{Deviation} = -0.022$; $W = 1.434 \times 10^9$; $p = 1.14 \times 10^{-8}$), indicating slight underperformance in the final match relative to the first. In contrast, medium first matches ($n = 81,867$) and hard first matches ($n = 78,592$) showed positive median changes (median $\Delta\text{Deviation} = +0.024$, $W = 1.586 \times 10^9$, $p = 3.16 \times 10^{-40}$; and median

$\Delta\text{Deviation} = +0.029$, $W = 1.432 \times 10^9$, $p = 4.83 \times 10^{-69}$, respectively), indicating a small improvement relative to SquashLevels expectations in later matches for these groups.

Figure 5 shows a violinplot of deviation scores grouped by match day demand, revealing easy first matches have a wider spread in deviation than harder matches though - perhaps indicating that easier/shorter matches lets random variation have a bigger effect as compared to harder/longer matches. Figure 6 illustrates the distribution of the change in deviation:

$$\Delta\text{Deviation} = \text{Deviation}_{\text{last}} - \text{Deviation}_{\text{first}} \quad (6)$$

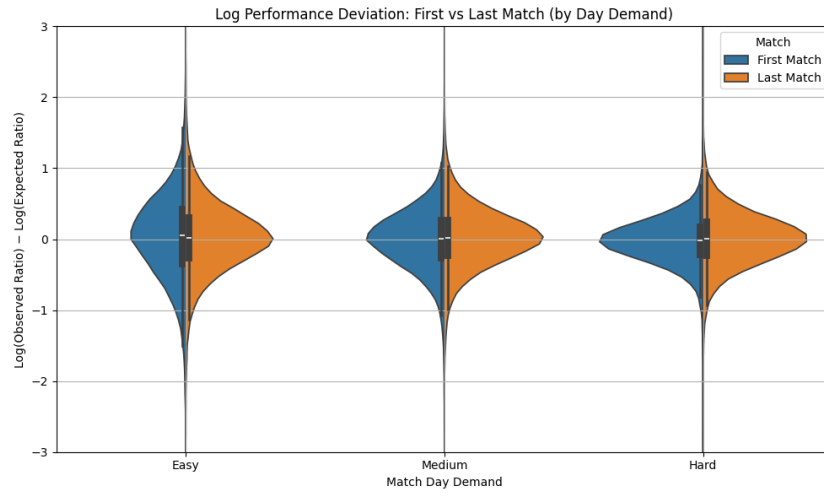


FIGURE 5: Deviation from expected performance in first and last matches of the day, grouped by match day demand. Vertical axis represents the value defined in Equation 5.

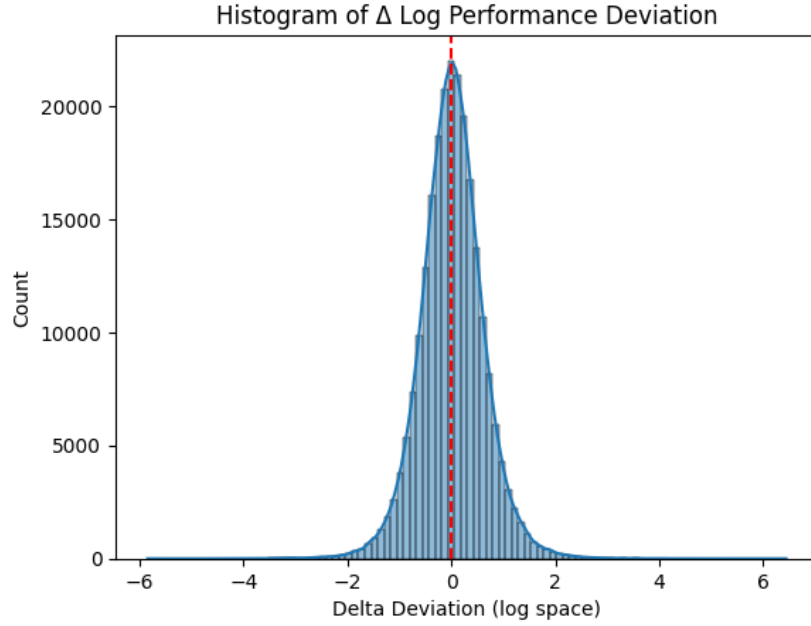


FIGURE 6: Histogram of Δ deviation (Equation 6) across 237,114 player-day pairs. Red vertical line marks the median.

These findings provide mixed evidence regarding fatigue-related underperformance across a day. When player-days were stratified by first-match demand, the easy-demand group showed a small but statistically significant negative shift in deviation, consistent with modest underperformance in the final match relative to the first. However, the medium- and high-demand groups exhibited small positive shifts, indicating that players marginally outperformed SquashLevels expectations in later matches under these conditions. This pattern suggests that any fatigue signal is not uniform and may be outweighed by warm-up effects, adaptation, or opponent/context variation, particularly when the first match is more demanding. From a rating system perspective, the results do not indicate a consistent, systematic downward bias in same-day rating-relevant performance, supporting the view that SquashLevels is broadly robust to intra-day performance shifts.

Effort Modulation in Mismatches

To investigate whether stronger players reduce effort in unbalanced matches, a residual-based analysis was conducted using the SquashLevels **RATIO** metric. This captures match dominance from the stronger player's perspective, combining both game and point scores.

A log-log linear regression model was fitted to estimate expected performance based on pre-match rating difference:

$$\log(\text{RATIO}) = \beta_0 + \beta_1 \cdot \log(\text{Level Ratio}) + \beta_2 \cdot [\log(\text{Level Ratio})]^2 + \epsilon \quad (7)$$

The model was highly statistically significant ($F(2, 3,826,048) = 8.386 \times 10^5$, $p < 0.001$) with an R^2 of 0.305, indicating that 30.5% of performance variation could be explained by rating gap. The fitted line was:

$$\log(\text{RATIO}) = 0.0115 + 0.8586 \cdot \log(\text{Level Ratio}) - 0.1272 \cdot [\log(\text{Level Ratio})]^2 \quad (8)$$

Residuals from the model were computed as:

$$\text{Residual} = \log(\text{RATIO}_{\text{observed}}) - \log(\text{RATIO}_{\text{predicted}}) \quad (9)$$

Matches were grouped into three bands based on SquashLevels conventions:

- **Fair:** Level Ratio 1.0–1.5
- **Moderate Mismatch:** Level Ratio 1.5–3.0
- **Extreme Mismatch:** Level Ratio > 3.0

Figure 7 displays the distribution of residuals across these bands. In Fair and Moderate and Extreme mismatches, median residuals were +0.006266, −0.024702 and −0.020966, respectively. The residual for Fair mismatch is near zero suggesting performance matched model expectations, but we see that in Moderate and Extreme mismatches the residuals are negative, indicating that stronger players underperformed relative to prediction.

This supports the hypothesis of *effort modulation*—that stronger players ease off in highly unbalanced matches—consistent with SquashLevels’ approach of reducing rating adjustment in these cases.

Figure 8 provides further context, plotting the fitted line over observed data. The plot shows an initially linear trend in the Fair mismatch regime that begins to flatten out as the Extreme mismatch regime is approached.

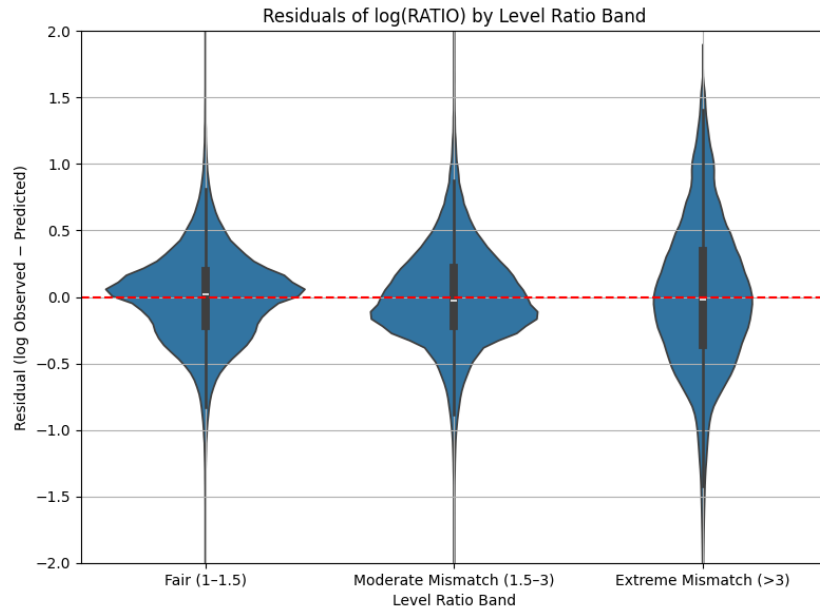


FIGURE 7: Residuals from Equation 9 grouped by level ratio band. The red dashed line indicates zero (perfect prediction).

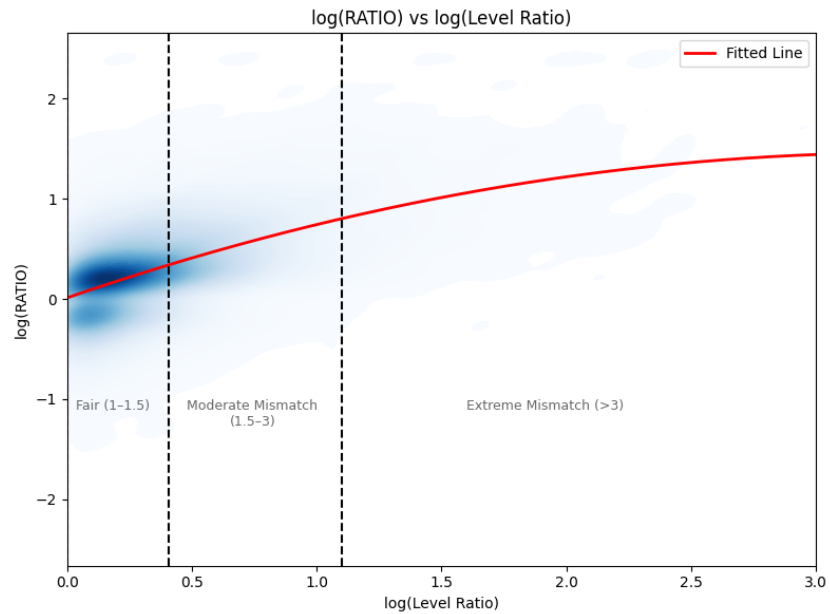


FIGURE 8: Heatmap of $\log(\text{RATIO})$ vs $\log(\text{Level Ratio})$ for stronger players. Red line is the fitted model (Equation 8).

Discussion

This chapter returns to the project’s central claim: that some observed match outcomes may deviate systematically from SquashLevels predictions due to *behaviour*, rather than genuine changes in underlying playing ability. The purpose here is not to restate the results in detail, but to interpret what the two analyses imply about (i) the plausibility of behavioural anomalies in this setting, (ii) how the two mechanisms relate to each other, and (iii) what the patterns suggest about SquashLevels’ modelling assumptions.

Fatigue Effects

The fatigue analysis was motivated by a straightforward mechanism: where a player competes multiple times in one day, accumulated physical and cognitive load could reduce later performance. If fatigue were a systematic behavioural anomaly, the final match of the day would be expected to deviate negatively from the SquashLevels expectation more often than the first match, and this effect would plausibly strengthen when the first match was more physically demanding.

The observed pattern provided mixed evidence and did not align with a simple fatigue-underperformance interpretation. When player-days were stratified by *first-match demand* (using points played in the first match as a proxy for initial load), the easy-demand group showed a small negative shift in deviation from first to last match, consistent with modest underperformance later in the day. However, the medium- and hard-demand groups exhibited small positive shifts, indicating slightly better-than-expected performance in the final match relative to the first. Rather than interpreting this as proof that fatigue does not exist in squash, the more defensible interpretation is that fatigue is not a dominant driver of rating-relative deviations under the conditions captured by this dataset and filtering, and that other same-day dynamics may outweigh any fatigue signal.

Several contextual explanations remain consistent with the findings. First, warm-up effects can plausibly offset or dominate fatigue at the day-level ordering used here: when

the first match is more demanding, players may reach peak movement efficiency, timing, and decision quality earlier in the day, improving later performance relative to expectation. Second, selection effects are likely: the subset of players who play multiple competitively weighted matches in a day may be fitter, more experienced, or participating in structured formats where match load is expected and managed. Third, although first-match points provide a stronger proxy than match order alone, the dataset does not encode key mediators of fatigue (e.g., rest time, match duration beyond total points, travel, or cumulative load across multiple days), so any fatigue signal may still be diluted when comparing only the first and last matches.

An important implication is methodological: day-level match ordering remains a coarse proxy for fatigue, and first-match points are only an indirect measure of physiological demand. The absence of a consistent negative shift—particularly in harder-demand groups—suggests that, within the current measurement resolution, SquashLevels’ assumption of stable within-day performance is not obviously violated in a way that would systematically bias ratings. In other words, the analysis provides limited support for fatigue as a behavioural anomaly of practical relevance in this data, while leaving open the possibility that more granular features (e.g., recovery time between matches) could reveal conditional fatigue effects.

Effort Modulation in Mismatches

Effort modulation was hypothesised to occur when a match is highly unbalanced. In such contexts the stronger player may reduce intensity once the outcome is effectively secure, producing scorelines that are less dominant than would be expected purely from the rating gap. Unlike fatigue, this mechanism is inherently contextual: it should be most visible in large skill disparities and may not appear at all in balanced matches.

The residual-based analysis supports this behavioural mechanism most clearly in the mismatch regimes beyond fair contests. Crucially, this inference depends on comparing observed dominance to an explicit baseline expectation. The regression residual framing therefore strengthens the behavioural interpretation: it isolates the question of whether performance is systematically *lower than predicted*, not merely whether stronger players tend to win (which is tautological in a rating system). Using a quadratic baseline relationship between $\log(\text{RATIO})$ and $\log(\text{Level Ratio})$ to capture the observed flattening at higher disparities, the residual analysis indicated that moderate and extreme mismatches were associated with negative median residuals. This pattern is consistent with reduced intensity, pacing, or social norm effects (e.g., avoiding excessive blowouts), and it aligns

with SquashLevels’ documented reasoning for dampening adjustments in highly unequal contests (SquashLevels, no date).

This result also highlights a key theoretical point: behavioural anomalies do not necessarily undermine a rating system; they often motivate the system’s practical mechanisms. In this case, the system’s mismatch handling can be interpreted as a behavioural correction rather than a purely statistical device. The analysis therefore offers empirical coherence between algorithm design and observed match behaviour, supporting the idea that highly unbalanced matches are less informative about true ability because behaviour changes systematically with context.

Integrating the two mechanisms

A useful way to integrate both findings is to distinguish between load-driven and context-driven behavioural effects. Fatigue is load-driven and depends on physiological depletion and recovery, which are not directly observed here beyond match ordering and a simple proxy based on first-match points. Effort modulation is context-driven and depends on incentives, perceived challenge, and the marginal value of additional points or games. The results suggest that, in this dataset, context-driven effects are more visible at scale than load-driven effects, and that behavioural deviations are most clearly detected when the match context directly changes incentives and effort.

This contrast also clarifies what “behavioural anomaly” means in practice for SquashLevels: not every plausible behavioural mechanism generates a consistent deviation in rating-relative outcomes, but certain match contexts (notably mismatches beyond fair contests) produce departures from baseline expectations that are consistent with behavioural adjustment. This distinction provides a coherent bridge into the next chapter, which focuses on what should be concluded, what can be recommended, and what limitations constrain those recommendations.

Conclusions and Further Work

Conclusions

Readers are already familiar with the two empirical patterns established in the Results chapter and interpreted in the Discussion: (i) fatigue effects did not produce a consistent same-day decline in rating-relative performance, and the direction of change depended on match-day demand as proxied by points played in the first match, and (ii) mismatch contests produced systematically lower-than-expected dominance by the stronger player when evaluated relative to an explicit baseline predictive model. This chapter consolidates what those patterns mean for the project’s unifying thesis and sets out the practical and theoretical value of the work, while also being explicit about limitations and priorities for future research.

The unifying thesis proposed that behavioural factors can create systematic deviations between observed match outcomes and those predicted by the SquashLevels model, without reflecting true changes in underlying playing ability. Based on the evidence in this dissertation, the thesis was partially accepted: behavioural anomalies were not universal across mechanisms, but they were observable and practically relevant in specific match contexts.

Two main conclusions followed. First, within competitively weighted matches and a first-versus-last match comparison, fatigue did not appear to introduce a uniform downward bias in rating-relative performance across a day. When player-days were stratified by first-match demand (easy/medium/hard based on `PTS_PER_MATCH` in the first match), the easy-demand group showed a small negative median change in deviation (median $\Delta\text{Deviation} = -0.022$, $n = 76,655$), consistent with modest underperformance in the final match. In contrast, the medium- and hard-demand groups showed small positive median changes (median $\Delta\text{Deviation} = +0.024$, $n = 81,867$; and median $\Delta\text{Deviation} = +0.029$, $n = 78,592$), indicating slight improvement relative to SquashLevels expectations in later matches for those groups. Given the very large sample sizes, these shifts were statistically significant, but their magnitudes were small, suggesting that any fatigue

signal is not consistent and may be outweighed by warm-up, adaptation, or match context effects.

Second, mismatch contests showed consistent underperformance by the stronger player relative to baseline model expectations, consistent with effort modulation. Using a quadratic log-log baseline to capture the observed flattening at higher level ratios, the model explained approximately 30.5% of variation in $\log(\text{RATIO})$ ($R^2 \approx 0.305$) over $n = 3,826,051$ matches. Median residuals were near zero in fair matches (approximately 0.006), but were negative in both moderate and extreme mismatches (approximately -0.025 and -0.021), indicating that stronger players won by a smaller margin than predicted as disparity increased. This provided clear evidence of a context-driven behavioural effect concentrated in mismatches beyond fair contests.

These findings led directly to practical recommendations for SquashLevels and similar ranking systems. Most importantly, mismatch dampening should be retained and viewed as a behavioural correction: the residual pattern in mismatches provides empirical justification for reduced sensitivity in rating updates when disparities are large, treating such matches as lower-information events and limiting distortion from low-intensity or socially moderated scorelines. By contrast, the fatigue results do not justify a generic same-day fatigue adjustment; the direction of the effect varies with first-match demand and the overall magnitude is small. Implementing a blanket correction without richer scheduling and recovery features could introduce bias where none is consistently demonstrated. More generally, behavioural corrections appear most defensible when targeted to match contexts where the deviation signal is strong and consistent, rather than applied uniformly.

Beyond practical guidance, the project contributed a clear modelling lesson: behavioural anomalies in an Elo-like ranking setting were more readily detected when tied to incentive structure and match context (effort modulation) than when they depended on unobserved mediators such as recovery time and physiological load (fatigue). This supports the broader principle that when behavioural mechanisms are hypothesised, the strongest empirical tests compare outcomes to an explicit baseline expectation and examine systematic residual structure rather than relying only on raw outcome differences.

Limitations and Further Work

Several limitations constrain interpretation and indicate where further work would be most valuable. For fatigue, the approach relied on a day-level comparison (first vs last match) and a proxy for initial load (points played in the first match). While this improves

on match-order alone, it still does not capture key mediators of fatigue such as recovery time between matches, match duration beyond total points, pacing within games, travel, or cumulative load across multiple days. As a result, this dissertation cannot claim that fatigue effects are absent in squash; rather, it found no evidence of a consistent fatigue-driven underperformance pattern under this proxy and filtering. A direct extension would incorporate richer fatigue proxies, such as time gaps between matches, duration proxies (e.g. total points played across the full day), and within-tournament sequences spanning multiple days.

For mismatch analysis, the baseline regression model was deliberately simple and used only the log level ratio (with a quadratic term) to model expected dominance. Residual patterns may partly reflect omitted contextual variables (e.g. scoring format, match weighting, and confidence). Future work should extend the baseline model with covariates such as **WEIGHTING**, confidence measures, and score type, then reassess residual shifts by mismatch severity. In addition, given the very large samples typical in this dataset, future reporting should place greater emphasis on practical effect sizes and uncertainty (e.g. bootstrap confidence intervals for median shifts) alongside significance tests.

Finally, a useful applied extension would be algorithm stress-testing: simulate rating trajectories with and without mismatch dampening to quantify downstream impacts on ranking stability and predictive accuracy. This would translate the behavioural findings into clearer guidance on whether current heuristics are near-optimal or could be improved.

In summary, this dissertation showed that behavioural anomalies in SquashLevels were context-dependent. Using simple, reproducible analyses, fatigue effects did not produce a consistent same-day underperformance signal and varied by first-match demand, while mismatch contests produced systematic underperformance relative to a quadratic baseline expectation, consistent with effort modulation. The key value of the thesis is that it validates a major behavioural assumption embedded in the SquashLevels algorithm (effective effort in mismatches) and provides evidence against introducing a generic fatigue correction without richer scheduling and recovery data.

Self-evaluation of Project Outcome

This project required me to work in a domain that was new to me: sports ranking systems and behavioural effects in performance data. Over the course of the dissertation, I developed a strong understanding of how Elo-like systems operate in practice and how SquashLevels adapts these ideas for squash. I also became more confident in reading academic literature, particularly in distinguishing between papers that propose new ranking models and those that evaluate existing systems and behavioural mechanisms. This improved my ability to extract defensible assumptions and position my work within published research.

A key strength of the project was that I maintained a scientific approach even when results did not support my initial expectations. In particular, the fatigue analysis did not show the expected underperformance, but I treated this as an informative outcome rather than forcing a narrative. I followed the scientific method as taught in the Research Methods in Data Science module: defining a testable hypothesis, selecting appropriate metrics, using statistical tests aligned with distributional assumptions, and interpreting results critically. I also kept structured notes throughout, which made it easier to resume work efficiently after interruptions and ensured consistency between the methodology, results, and final write-up.

In terms of process, I applied project management principles from the same module to organise tasks, plan the structure of the dissertation early, and track progress against milestones. This helped me stay focused and avoid scope creep, especially once I decided to narrow the dissertation to two core behavioural mechanisms.

If I were to repeat the project, there are two main areas I would improve. First, although I used project management methods, I struggled to maintain the original timeline when project requirements changed or unfavourable external factors emerged. In future, I would liaise more proactively with my supervisor to re-plan deliverables earlier and reduce last-minute restructuring. Second, I found it challenging to judge the appropriate depth for each chapter. Next time, I would request access to high-quality past dissertations

(where available) to calibrate expectations and strengthen my writing plan from the start.

Appendix A

GenAI Usage Statement

Generative AI tools were used to support drafting and language editing, and to assist with producing and refining analysis code. All outputs were reviewed, edited, and verified by the author. No personally identifiable information was provided to any GenAI tool.

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